Exam.Code:0506
Sub. Code: 3881

2053

Master of Computer Application

Second Semester

CS-2115: Object Oriented Programming (Through C++ & Java)

Time allowed: 3 Hours Max. Marks: 80

NOTE: *Attempt five questions in all, including Question No. IX (Unit-V) which is compulsory and selecting one question each from Unit I -IV.*

x-x-x

UNIT – I

I. a) Explain various characteristics of OOPs in detail.

b) Explain various operators and variables through OOPs program. (2x8)

II. a) What is array? Explain the use of multidimensional array through C++ program.

b) Define string. Explain any three string handing functions. (2x8)

UNIT - II

III. a) Define inheritance. How are various modes accessed through this features? Explain.

b) What is File? Explain and program various file handling functions. (2x8)

IV. a) What is function and scope of variables within a function? Explain and demonstrate Pass-by-value and Pass-by-reference in this context.

b) What is polymorphism? Explain and implement dynamic polymorphism. (2x8)

UNIT - III

V. a) What are JVM and byte code? Also explain some other important characteristics of Java language.

b) Why are interface used? Implement interfaces through function. (2x8)

VI. a) Draw difference between overloading and overriding through program and well explain the code functionality.

b) What is package? How are they imported? Explain. (2x8)

P.T.O.

UNIT - IV

VII. a) What is thread? How is it crated,implemented and extended? Demonstrate.

b) Define and implement few of AWT controls. (2x8)

VIII. a) How does a thread go through various stages of its life-cycle? Exemplify.

b) What is applet? How are they crated and used through applet class? Exemplify. (2x8)

UNIT - V

IX. Explain the following:-

(a) Expression

(b) Tokens

(c) Templates

(d) Exception handling

(e) Data-types of Java

(f) Inheritance

(g) Synchronization

(h) Layout manger (8x2)

x-x-x

Exam.Code:0506
Sub. Code: 3882

2053

Master of Computer Application

Second Semester

CS-2116: Computer Networks and Security

Time allowed: 3 Hours

Max. Marks: 80

NOTE: Attempt five questions in all, including Question No. 9 (Section-E) which is compulsory and selecting one question each from Section A-D.

x-x-x

SECTION-A

1) Define and differentiate the following pairs:
   a) Wired and wireless networks
   b) OSI and TCP/IP network reference models (8, 8)

2)
   a) In what way is link state routing better than Distance Vector Routing. Explain in detail.
   b) What are the reasons of congestion in networks? Describe the general principle of congestion control? (8, 8)

SECTION-B

3)
   a) What do you mean by connection-oriented protocol? Explain with example. What are the features of TCP for which it is called as connection-oriented protocol?
   b) What is difference between TCP and UDP protocol? Explain with diagrams. (8, 8)

4)
   a) Describe how email works. Describe the key components and flows. Identify key standards that apply. Use figures as needed.
   b) Distinguish between the http:// and https:// protocols. What is the main advantage of persistent connections in HTTP? (8, 8)

SECTION-C

5) What is cryptography? Differentiate between public-key and private-key cryptography. Explain the basic principles of public key cryptography. Explain the working of RSA algorithm with a suitable example. (16)

6)
   a) What is a digital signature? How does it work? Explain how digital signatures are used for message-by-message authentication.
   b) What is e-mail security? How is email security implemented? Where do we do e-mail filtering? Explain. (8, 8)

SECTION-D

7) What is a firewall in a network? Explain the difference between restrictive and permissive policy as two philosophies of a firewall. Give example of Firewall Rules for deploying a firewall at the network perimeter. (16)

8)
   a) What is Intrusion Detection System (IDS)? Explain the difference between Host IDS and Network IDS with examples.
   b) What does PGP stand for? What is its purpose? What are the five principle services provided by the PGP? (8, 8)

SECTION-E

9)
   a) You have different user accounts in different applications. How do you manage different passwords? Suppose you forgot the password, then how do you recover it?
   b) What are four main goals of secure network?
   c) Name two cryptographic techniques used by Transport Layer Security (TLS).
   d) Give five modes of operation of Block Cipher.
   e) Differentiate the terms unicast, broadcast and multicast communication.
   f) What is IP addressing? How is it classified?
   g) For n devices in a network, what is the number of cable links required for a mesh and star topology?
   h) What are the components of a data communication system? (8x2=16)

x-x-x

Exam.Code:0506
Sub. Code: 3883

2053

Master of Computer Application

Second Semester

CS-2117: Software Engineering and Project Management

Time allowed: 3 Hours

Max. Marks: 80

NOTE: *Attempt five questions in all, including Question No. IX (Unit-V) which is compulsory and selecting one question each from Unit I -IV.*

x-x-x

UNIT – I

I. a) What is process model? Compare waterfall and spiral.

b) How is COCOMO implemented to estimate efforts? Exemplify. (2x8)

II. a) What is SRS? Explain various metrics for requirement specification quality.

b) What is project management and how is it carried out and why? Explain. (2x8)

UNIT - II

III. a) What is the importance of project scheduling? Develop a project N/W diagram for any project.

b) Define S/W quality and its features. Which are the various factors to affect it? (2x8)

IV. a) How is structured analysis carried out? Explain.

b) How do you build SQAP? Exemplify the procedure. (2x8)

UNIT - III

V. a) How is software testing process carried out? List out any two testing strategies.

b) Compare black-box and white box testing techniques. (2x8)

VI. a) What is software testing? Why and how is static and dynamic analysis carried out for it? Explain.

b) Explain the various four techniques of white box testing. (2x8)

P.T.O.

UNIT - IV

VII. a) Define S/w maintenance along with its types in detail.

b) Why is SCM required for developing successful project? How one can plan and organize for SCM? (2x8)

VIII. a) Explain various software maintenance process models.

b) Illustrate the features of 'version and release management'. (2x8)

UNIT - V

IX. Explain the following:-

(a) SCM principles

(b) Objectives of S/w testing

(c) Debugging

(d) Critical path analysis

(e) ISO 9000

(f) Release management

(g) LOC

(h) S/w inspection (8x2)

x-x-x

Exam.Code:0506
Sub. Code: 3884

2053

Master of Computer Application

Second Semester

CS-2118: Web Technologies and Python Programming

Time allowed: 3 Hours

Max. Marks: 80

NOTE: *Attempt five questions in all, including Question No. IX (Unit-V) which is compulsory and selecting one question each from Unit I - IV.*

x-x-x

UNIT - I

I. What do you mean by pattern matching? Write a program in JavaScript to illustrate the concept of pattern matching. (16)

II. What is List? How can you create different types of lists in HTML? (16)

UNIT - II

III. What do you mean by AJAX? Describe the features of AJAX. (16)

IV. What do you mean by Angular JS? Explain the MVC architecture in detail. (16)

UNIT - III

V. What is String in Python? Explain different types of string related functions. (16)

VI. What do you mean by conditional statement? Explain different types of conditional statements available in Python. (16)

UNIT - IV

VII. What is a module? How can you create and access modules in Python? (16)

VIII. What is Django? How can you create web pages using Django? (16)

UNIT - V

IX. Write short notes on following:-

a) File handling in Python

b) List slicing in Python

c) Cascading style sheets

d) XML

(4x4)

x-x-x

Exam. Code: 0506
Sub. Code: 3885

2063

Master in Computer Applications (MCA)-2nd Semester

CS-2119: Analysis and Design of Algorithms

Time allowed: 3 Hours | Max. Marks: 80

NOTE: *Attempt five questions in all including Q. No.-IX (Unit-V) which is compulsory and selecting one question each from Unit I-IV.*

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UNIT-I

I. (a) How is an algorithm analysed? Exemplify its complexity rotations in detail.

(b) What is devide and conquer practice? Exemplify through any method of your choice. (8+8)

II. (a) What is the importance of recurrence relations? Solve any using repeated substitutions.

(b) Draw difference between merge and quick sort techniques in detail. (8+8)

UNIT-II

III. (a) What is minimum spanning tree? Analyze it through example.

(b) How are optimal binary search trees programmed to solve any problem? Explain. (8+8)

IV. (a) How is "Knapsack problem" solved using greedy approach? Explain.

(b) Explain and exemplify "Travelling Salesman Problem". (8+8)

UNIT-III

V. (a) How are Hamiltonian cycles located through back-tracking? Exemplify.

(b) What is "Branch-and-Bound" technique? Exemplify O/I Knapsack to implement it. (8+8)

VI. (a) Which are the various issues for "Back Tracking"? Exemplify through Graph-Coloring.

(b) Solve O/I Knapsack through "Branch-and Bound" method. (8+8)

UNIT-IV

VII. (a) What is "Lower Bound" & its significance? Exemplify them for "Parallel Computations"?

Contd.....P/2

(b) How are NP-Complete problems solved? Justify through any example. (8+8)

VIII. (a) Apply "Lower-Bound" theory for "Ordered Searching and Sorting" way.

(b) Draw difference between deterministic and non-deterministic algorithms. (8+8)

UNIT-V

IX Explain: -

(a) NP-Hard problems

(b) Cook's Theorem

(c) 8-queen's problem

(d) Recursive algorithms

(e) Analysis and design of algorithms

(f) Strassen's matrix

(g) Job-sequencing problem

(h) Greedy analysis (8×2)

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Exam.Code:0506
Sub. Code: 3882

2123

Master of Computer Applications

Second Semester

CS-2116: Computer Networks and Security

Time allowed: 3 Hours | Max. Marks: 80

NOTE: Attempt five questions in all, including Question No. 9 (Section-E) which is compulsory and selecting one question each from Section A-D.

x-x-x

SECTION-A

1) Define and differentiate the following pairs:
   a) IPv4 and IPv6 network addressing
   b) Physical network layer and data link network layer (8, 8)

2)
   a) What is Routing? Discuss various routing algorithms. Write the distance vector routing algorithm to compute a minimum cost among the nodes with the help of an example.
   b) What are general principles of congestion control? Explain Leaky Bucket Algorithm. (8, 8)

SECTION-B

3)
   a) The Internet uses both TCP and UDP protocols to deliver services to the users. Identifying a service delivered by each protocol, discuss the characteristics of each protocol.
   b) Discuss the following question related to the transport layer:
      i) Many features of the transport layer (e.g., flow control and reliable delivery) are also handled by the data link layer. Is this duplication of effort? Why or why not?
      ii) Discuss the problems with 2-way handshake in connection establishment and connection termination? How does 3-way handshake resolve it? Explain through diagrams. (8, 8)

4)
   a) What is the essence of DNS (Domain Name system)? How does it map to IP address? What is the role of DNS when sending mail to amit@pu.ac.in? Be precise.
   b) Describe how WWW works. Describe its key components. Identify key standards that apply.(8, 8)

SECTION-C

5) What is cryptography? State and explain the principles of public key cryptography. Explain the working of RSA algorithm with example. Describe the key management in RSA cryptography algorithm in detail. (16)

6)
   a) Define the main principles of network security. Discuss different types of attacks on network.
   b) What is a digital signature? How does it work? Illustrate how a hash code is used to provide digital signature? (8, 8)

SECTION-D

7) Define network firewall. Explain the architecture of a firewall. Which is generally safer (from a security point of view), a firewall with a "default deny" policy or a firewall with a "default allow" policy? Why? (16)

8)
   a) What is web security? What are web security requirements? Illustrate SSL/TLS operations in detail.
   b) What is IP Security (IPSec)? How does IPSec work? Explain. (8, 8)

SECTION-E

9)
   a) Distinguish between plain text and cipher text.
   b) What is the significance of Secure Electronic Transaction (SET)?
   c) Describe two fundamentally different conceptual approaches that can be used for user authentication.
   d) Differentiate between symmetric and asymmetric encryption.
   e) What do you mean by wireless transmission? Briefly describe the various media that support wireless transmission.
   f) What do you mean by baud rate and bit rate? Is there any relation between these two?
   g) A network has a subnet mask *255.255.252.0*. How many hosts can this network have?
   h) What are the fundamental measures of interest for a data communications system? (8x2=16)

x-x-x

Exam.Code:0506
Sub. Code: 3884

2123

Master of Computer Applications

Second Semester

CS-2118: Web Technologies and Python Programming

Time allowed: 3 Hours | Max. Marks: 80

NOTE: Attempt five questions in all, including Question No. IX (Unit-V) which is compulsory and selecting one question each from Unit I - IV.

x-x-x

UNIT - I

I. What do you mean by constructor? Write a program in JavaScript to illustrate the concept of constructor. (16)

II. What is Cascading Style sheet? Compare and contrast different types of cascading style sheets with examples. (16)

UNIT - II

III. What do you mean by XML? Describe the features of XML. (16)

IV. What do you mean by AJAX? How can you handle Dynamic HTML with AJAX? (16)

UNIT - III

V. What is List in Python? Explain different types of list related functions. (16)

VI. What do you mean by control statement? Explain different types of loops available in Python. (16)

UNIT - IV

VII. What is a function? Write a program in C to generate Fibonacci series using function in Python. (16)

VIII. What is file? Explain the different types of file related functions in Python. (16)

UNIT - V

IX. Write short notes on following:

a) Django framework

b) Strings in Python

c) Angular JS

d) DOM2 (4x4)

x-x-x

Exam.Code:0506
Sub. Code: 3885

2123

Master of Computer Applications
Second Semester
CS-2119: Analysis and Design of Algorithms

Time allowed: 3 Hours **Max. Marks: 80**

NOTE: Attempt five questions in all, including Question No. IX (Unit-V) which is compulsory and selecting one question each from Unit I - IV.

x-x-x

UNIT-I

1 a) How to find time and space complexity of an algorithm? Explain with the help of an example. (8)

b) Write an algorithm to sort an array using Quick Sort. Discuss best and worst case time complexity of the algorithm. (8)

2 a) What are Asymptotic notations? Explain with the help of suitable examples. (8)

b) How to multiply two matrices using divide and conquer technique? What is the time complexity of the method? (8)

UNIT-II

3 a) Define Job Sequencing with deadlines problem. Give an example. How to solve it using greedy technique? (8)

b) Solve the following instance of 0/1 Knapsack problem using dynamic programming technique:
n = 3; (w1, w2, w3) = (3, 3, 4); (p1, p2, p3) = (2, 4, 6); capacity of knapsack(m) = 8 (8)

4 a) Explain Prim's algorithm to find Minimum Cost Spanning tree. Explain its complexity. (8)

b) What is Optimal Binary Search Tree problem? Write an algorithm using dynamic programming technique to solve this problem. (8)

UNIT-III

5 a) Discuss Backtracking technique with the help of a recursive algorithm. (8)

b) What do you understand by Least Cost (LC) search? Take an instance of 15-puzzle problem and solve it using LC search. (8)

6 a) What is Hamiltonian Cycle problem? Write an algorithm to solve this problem using backtracking. (8)

b) How to solve Travelling Salesperson problem using Least Cost search? Explain with the help of an example. (8)

UNIT-IV

7 a) What is Parallel Comparison Tree (PCT)? Draw a PCT with two processors to sort three numbers. (8)

b) What are non-deterministic algorithms? Write a non-deterministic algorithm to solve 0/1 Knapsack problem. (8)

8 a) What are NP-Hard problems? How to prove a problem NP-Hard? Give two examples of NP hard problems. (8)

b) What are different methods to evaluate a polynomial? What is the lower bound on number of additions/subtractions for evaluating an nth degree polynomial. (8)

P.T.O.

UNIT - V

9	a)	Give important properties of an algorithm.	(2)
	b)	Discuss the complexity of Binary Search.	(2)
	c)	How is Kruskal's technique different from Prim's technique?	(2)
	d)	Define Principle of Optimality. Give example.	(2)
	e)	Define live node and dead node with the help of an example.	(2)
	f)	What is D-search? Give example.	(2)
	g)	When is a parallel algorithm work optimal?	(2)
	h)	Define Halting problem.	(2)

x-x-x